

Moment compression of visited-range distributions for extensive-lesion robustness in layered Markov chains

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A layered feed-forward Markov chain reaches an absorbing target with probability R_0 ; deleting an extensive, random set of spatial regions lowers it, and the strength η at which the survival collapses is a natural robustness measure. Framing this as a coarse-graining question—how much microscopic trajectory information must be retained to predict the macroscopic response—we study a registered family of doubly stochastic reactive kernels whose one-time statistics are matched by construction. Fixing the intervention to persistent, dead-end spatial lesions, we prove that the disorder-averaged survival is an exact functional of a *single* scalar random variable: V , the number of distinct coarse-grained regions a trajectory visits. The whole robustness curve is $\mathbb{E}_\omega[(\binom{n_b-V}{r})/(\binom{n_b}{r})]$. This reduces the original “which statistic predicts robustness” question to a well-posed compression problem: how many moments of the law $P(V)$ are needed to reproduce the curve? Across the family the mean alone is insufficient in most cases; the mean and variance together reproduce the curve to 1% in 105 of 120 cases; higher-order information remains necessary in a localized set of cases, namely coarse-resolution random permutations and sparse block kernels. The study was fully preregistered, frozen before any outcome was computed, and independently re-implemented.

I. INTRODUCTION

Many high-dimensional stochastic systems display comparatively simple macroscopic behavior despite substantial heterogeneity in their microscopic dynamics. This raises a basic coarse-graining question: how much microscopic information must be retained to predict a chosen macroscopic response? A particularly strong form of coarse graining would hold if that response were fixed by a scalar descriptor, or by only a few low-order statistics of the microscopic ensemble. Whether such a reduction is possible, and how far it can be pushed before the response is lost, is generally unclear and depends on both the system and the response one asks about.

We turn this into a controlled test problem. Take the response to be the robustness of stochastic transport under an *extensive* perturbation—the removal of a finite fraction of the state space—which cannot be captured by any single-time statistic and so forces the question of what microscopic information actually matters. Concretely we study a feed-forward (layered, non-revisiting) Markov chain with an absorbing target, and remove a random fraction of coarse-grained spatial regions for the whole duration, in “dead-end” mode (a trajectory entering a removed region fails). This is the setting in which the folk intuition that “many parallel routes” confer robustness would live; we keep that intuition only as one physical instantiation of the coarse-graining question, not as the motivation, and we deliberately do *not* assume a scalar descriptor at the outset.

The analysis then proceeds as two stages of coarse graining. First, an *exact* one: for this intervention the disorder-averaged robustness curve is a linear functional of a single one-dimensional law, the distribution $P(V)$ of the number of distinct regions a trajectory visits (its *range*). The full microscopic trajectory ensemble is thus

reduced, without approximation, to $P(V)$. Second, an *approximate* one that we can now pose sharply: since $P(V)$ is exactly sufficient, how much further can it be compressed—to its mean, to its mean and variance, or not at all—while preserving the macroscopic robustness curve? This replaces the ill-posed question “which trajectory statistic predicts robustness” by the well-posed question of how far the sufficient distribution compresses. Across a preregistered family of kernels we then *map* the regimes: the mean is insufficient in most tested regimes, the mean and variance suffice to a preregistered tolerance in the large majority of cases, and the exceptions are not scattered but localized.

a. Relation to prior work. That survival under trapping is controlled by the range—the number of distinct sites visited—is classical. The range of a random walk is a much-studied observable [1, 2], and for sites that are independently absorbing traps at concentration c the survival probability is the range functional $\Phi = \mathbb{E}[(1-c)^V]$, whose mean-range approximation is the Rosenstock estimate [3]; trapping and survival on structured substrates remain an active subject [4]. Because $\Phi = \mathbb{E}[e^{-\lambda V}]$ with $\lambda = -\ln(1-c)$, its logarithm is the cumulant generating function of the range and the Rosenstock estimate is its leading term, so mean-, variance-, and higher-order approximations to trapping survival correspond to a cumulant expansion whose corrections have been studied [2, 4]. We therefore do not claim to discover either that survival is a functional of the range or that its moments enter in an order. Our identity, Eq. (1), is the finite-system, sampling-without-replacement counterpart of Φ : a fixed number r of coarse-grained regions is removed persistently, giving a hypergeometric rather than an independent-Bernoulli weight. The question here is complementary: given a range law that is *exactly* sufficient for a finite extensive lesion, how far can it be compressed—to how many moments—while preserving

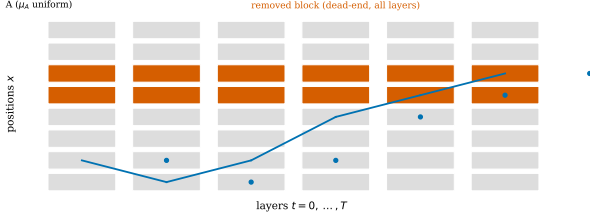


FIG. 1. Layered chain and intervention (schematic). Positions x at each layer $t = 0, \dots, T$; the initial layer A carries a uniform reactive marginal. Regions (blocks of ℓ consecutive positions) are removed for all layers (persistent), and in dead-end mode a trajectory entering a removed block fails. A surviving trajectory (blue) avoids the removed block at every layer.

the *entire* response curve $R(\eta)$ to a prescribed accuracy, across a controlled family of dynamics. What is new is thus not the range dependence but the use of the exact reduction as the first stage of a coarse-graining hierarchy—microscopic ensemble $\rightarrow P(V) \rightarrow$ low-order moments—and the preregistered map of where each level of that hierarchy suffices to reproduce the robustness response. Robustness to region removal is also studied in network percolation [5], a different setting from the matched-family, fixed-dynamics comparison made here.

The work was carried out as a preregistered study [10]: the model family, the intervention, the closures, the decision statistic, the threshold, and the kill criteria were fixed and hash-frozen before any robustness curve was computed, and the analysis pipeline was independently re-implemented. We describe this machinery briefly here and in full in the Appendix; the scientific content is the identity of Sec. III and the compression result of Sec. V, and the preregistration serves reliability, not novelty.

II. MODEL AND INTERVENTION

The state space (Fig. 1) has n positions per layer and $T+1$ layers $t = 0, \dots, T$; after layer T an absorbing target B is reached, and an absorbing failure state F is accessible from every layer. The initial layer A carries a uniform reactive marginal $\mu_A(x) = 1/n$. Dynamics are specified reactively [9]: we choose doubly stochastic kernels P_t^R and embed them by a uniform leak, $P_t(x, y) = (1 - \lambda)P_t^R(x, y)$ and $P_t(x, F) = \lambda$. The committor is then $q_t(x) = (1 - \lambda)^{T-t}$ (flat within each layer), the overall reactive probability is $R_0 = (1 - \lambda)^T$, and the Doob-conditioned process is exactly P_t^R . Uniform reactive marginals make all one-time (single-layer) statistics identical across the family by construction, which is the matching we want.

We coarse-grain positions into $n_b = n/\ell$ blocks of ℓ consecutive positions and index them $b = \lfloor x/\ell \rfloor$. An extensive lesion of strength η removes a uniformly random set D of $r = \lceil \eta n_b \rceil$ blocks, the same blocks at every layer

(persistent). In dead-end mode the lesioned transfer operator is the restriction of P^R to the surviving columns.

The registered family (Appendix A) has 27 time-homogeneous members drawn from five menus—localized walks $\text{Loc}(\sigma)$, permutations $\text{Perm}(\pi)$, mixing kernels $\text{Mix}(c)$, block kernels $\text{Block}(M, c)$, and shifted-mixing kernels $\text{ShiftMix}(c)$ —at $n = 256$, $T = 64$, $\lambda = 1 - 2^{-1/64}$ so that $R_0 = 1/2$, and resolutions $\ell \in \{1, 2, 4, 8, 16\}$.

III. ROBUSTNESS IS A FUNCTIONAL OF THE VISITED-RANGE LAW

Let $S = D^c$ be the surviving blocks and, for a trajectory ω , let $V(\omega)$ be the number of *distinct* blocks it visits over layers $0, \dots, T$. In dead-end mode a trajectory survives the lesion D iff it never enters a removed block, i.e. iff its visited set avoids D . Averaging the survival over uniformly random removal sets of size r and using that a fixed visited set of size V avoids D with probability $\binom{n_b - V}{r} / \binom{n_b}{r}$ gives the central identity

$$\frac{\mathbb{E}_D[R_{\text{dead}}(\eta)]}{R_0} = \mathbb{E}_\omega \left[\frac{\binom{n_b - V(\omega)}{r}}{\binom{n_b}{r}} \right], \quad r = \lceil \eta n_b \rceil. \quad (1)$$

The disorder average has been carried out in closed form; the right-hand side is an expectation over trajectories only. Thus, *for this intervention, the entire robustness curve depends on the process solely through the one-dimensional law $P(V)$* —not through any finer trajectory or network statistic. The identity is elementary: it restates survival as the avoidance of a random set and averages that event in closed form. That simplicity is a feature, not a limitation—it pins the sufficient statistic exactly, with no modeling freedom, and it is what turns a vague “which statistic matters” question into the sharp compression problem below. Equation (1) is verified to machine precision on a small exactly enumerable chain (Appendix B) and, on the registered family, by direct removal-set simulation (Sec. VI).

Two remarks fix the scope. First, the reduction is special to dead-end mode. In “redirect” mode (mass entering a removed block is renormalized onto surviving successors) the row-sum normalization preserves the within-layer flatness of the committor whenever no row is fully removed, so $R_{\text{redirect}}(\eta)/R_0 = |S|/n$ is *kernel-independent*; a dead row can occur only if a state’s entire support lies in D , which is impossible when the diagonal is positive. Redirect therefore carries no family information for most members and is not the informative intervention (Appendix B). Second, Eq. (1) is a statement about feed-forward dynamics, persistent lesions, dead-end mode, and the uniform embedding; we make no claim beyond that class.

IV. THE COMPRESSION QUESTION AND CLOSURES

Equation (1) recasts the original question. Instead of asking which graph statistic predicts robustness, we ask how much of the sufficient law $P(V)$ is needed to reproduce the curve. This defines an information hierarchy

$$\mathbb{E}[V] \subset \{\mathbb{E}[V], \text{Var}(V)\} \subset P(V). \quad (2)$$

We test the two lower rungs with maximum-entropy closures on the support $v = 1, \dots, n_b$: the mean-only closure $p_1(v) \propto e^{\lambda_1 v}$ and the mean-and-variance closure $p_2(v) \propto e^{\lambda_1 v + \lambda_2 v^2}$, with the Lagrange multipliers fixed by the exact moments $\mathbb{E}[V]$ and $\text{Var}(V)$ (both computed exactly by block-avoidance recursions). The two closures are built by the same principle, so the only difference between the rungs is the amount of information supplied.

Substituting a distribution p into the weight of Eq. (1) yields a curve $R_p(\eta)/R_0 = \sum_v p(v) w(v, r(\eta))$ with $w(v, r) = \binom{n_b - v}{r} / \binom{n_b}{r}$. Writing R^* for the true curve (from the exact $P(V)$ where available, otherwise from a disorder-integrated trajectory estimator that removes removal-set noise by construction), the registered decision statistic is, for each member j , the sup-norm curve distance

$$D_{k,j}(\ell) = \sup_{\eta} \frac{|R_{k,j}(\eta) - R_j^*(\eta)|}{R_0}, \quad k = 1, 2, \quad (3)$$

evaluated on a fixed geometric η -grid that brackets the analytic collapse limits. A cell is counted as “mean-insufficient” when $D_{1,j}(\ell) > \tau$ with the registered tolerance $\tau = 10^{-2}$; a family verdict follows an existential rule over members (Appendix D).

V. RESULTS: THE REGIME MAP

Figure 2 is the central result. Of the 135 (member, resolution) cells, 15 are mean-sufficient: these are the degenerate cases where the visited count is a single value or sits at a support endpoint (the identity permutation at all ℓ ; shifts whose orbit fills the blocks; the near-saturated ShiftMix at coarse blocks). In the remaining 120 cells the mean alone fails the tolerance, so the family verdict is that the mean is *not* a sufficient coordinate. This verdict is a coarse gate—it is essentially fixed by the presence of concentrated members whose visited count is not captured by its mean—and the informative object is the regime map itself, to which we now turn.

The compression result is what happens at second order. In 105 of those 120 cells—88%—the mean-and-variance closure reproduces the entire robustness curve to within $\tau = 10^{-2}$: two numbers suffice. This regime covers all of the localized and mixing families (Fig. 2a, blue) and is illustrated in Fig. 3a, where the closure-2 curve lies on the truth while the mean-only curve is visibly biased.

The exceptions are structured, not scattered. The 15 non-closing cells comprise 14 in which the second-order closure improves on the mean-only closure but remains above threshold, and one higher-order cell (where the second moment does not substantially help, at the threshold). They are exactly the random permutations at coarse resolution and the sparse block kernels $\text{Block}(\cdot, 0.02)$; the single higher-order cell is $\text{Perm}(\text{random-2})$ at $\ell = 8$, with $D_1 = 0.012$ just above τ . These cells coincide with broad visited-count distributions—for example $\mathbb{E}[V] \approx 56$ with $\text{Var}(V) \approx 3.7 \times 10^2$ for a random permutation at $\ell = 1$ —but the mechanism underlying this localization is not tested here (see Sec. VII). A representative non-closing curve is shown in Fig. 3b: closure-2 improves on closure-1 but the truth is not yet captured.

a. Sensitivity to the tolerance. The count 105/120 is reported at the registered $\tau = 10^{-2}$, fixed before any curve was computed. Re-tabulating the same frozen distances at other thresholds (no new computation) shows the picture is not an artifact of that choice: the fraction of exceeding cells closed by second order rises smoothly through 67%, 78%, 88%, 96% at $\tau = 3 \times 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 2 \times 10^{-2}$, and the non-closing set stays the same members (coarse-resolution random permutations and sparse blocks) throughout. Only at the loosest $\tau = 5 \times 10^{-2}$ does it vanish. The qualitative result—mean insufficient, second order dominant, a residual non-closing set—is stable across a decade of τ .

b. Scalar coherence does not determine the sufficient coordinate. As a strong control, kernels sharing an identical scalar path coherence $C = \sum_t I(X_t; X_{t+1})$ nonetheless produce robustness curves that differ by up to 0.86 in sup norm (e.g. the identity, a shift, and a random permutation at $\ell = 1$ all have $C = T \ln n$ yet differ maximally). A single scalar coherence therefore does not fix the low-order description that Eq. (1) makes operative.

VI. CONTROLS AND VALIDATION

Under *layer-independent* removal (each block removed independently at each layer) the dead-end survival is $(1 - \eta)^{T+1}$ for every member, independent of the kernel; the visited-range identity plays no role and all members coincide. This confirms that the persistent structure is what makes robustness carry family information.

Direct removal-set simulation (200 random lesions per point) agrees with the disorder-integrated estimator within sampling noise; for the fully exchangeable Mix kernels the survival is identical for every removal set of a given size, so the direct simulation matches the estimator to machine precision, an exact check of Eq. (1). The pipeline passed nine registered code checks, including an oracle validation that compared exact $P(V)$ curves against the estimator and caught a catastrophic-cancellation failure in one closed-form route (corrected by a stable occupancy recursion). An independent re-

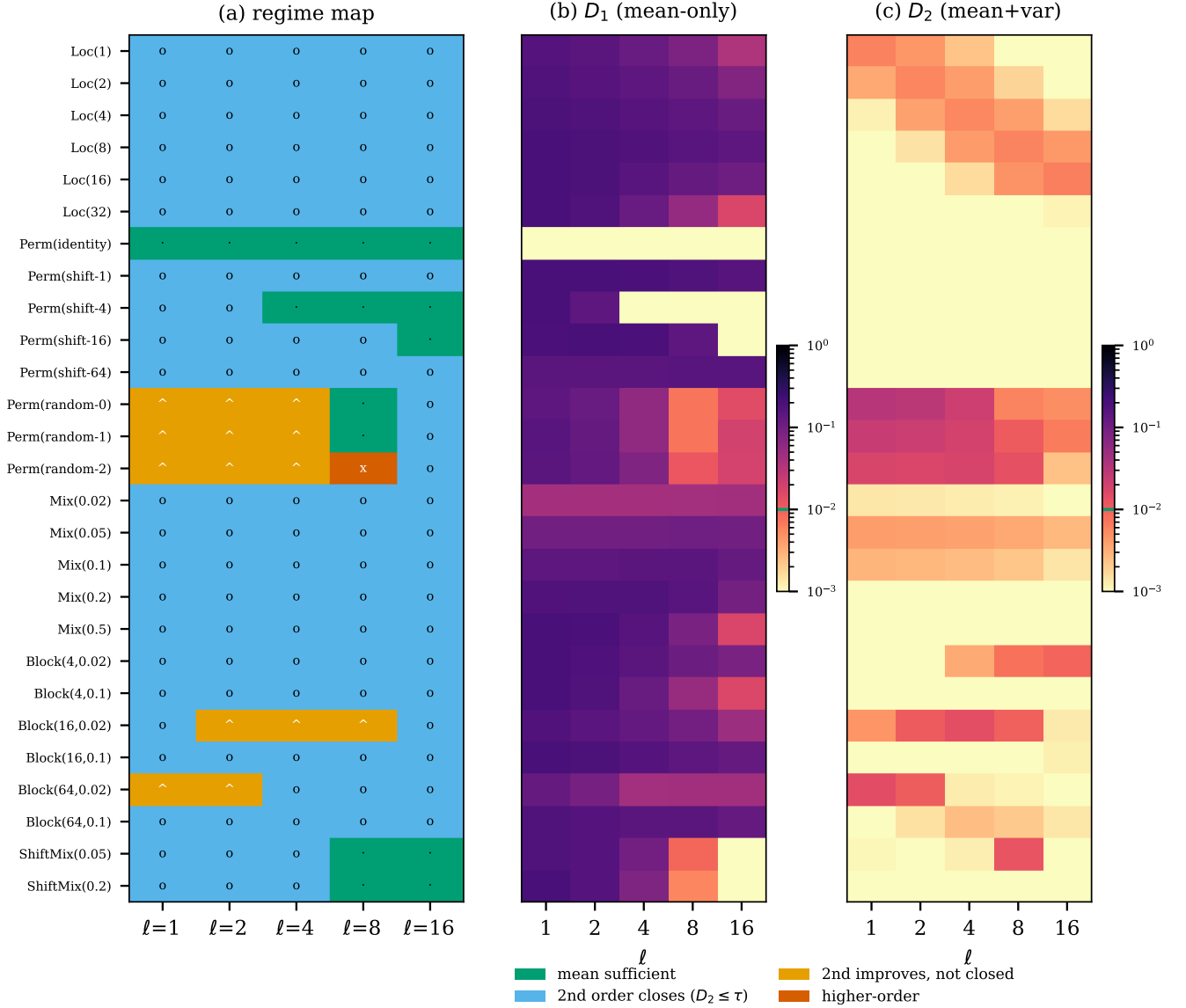


FIG. 2. Regime map over the 27 members ($\times$ five resolutions ℓ). (a) Categorical outcome per cell: mean sufficient (green), mean+variance closes the curve to $\tau = 10^{-2}$ (blue), mean+variance improves but does not close (orange), higher-order required (red). Glyphs repeat the category for print/CVD safety. (b),(c) The underlying distances D_1 (mean-only) and D_2 (mean+variance) on a log scale; the green line on each colorbar marks τ . The mean is dark (insufficient) almost everywhere in (b); adding the variance drives D_2 below τ (pale) except on the random-permutation and sparse-block rows.

implementation of the entire pipeline from the frozen family specification reproduced the moments and curves within the registered tolerances.

VII. DISCUSSION

For the class studied here, robustness to extensive spatial lesions is not a property of the full trajectory network but of a single scalar law, the visited-range distribution $P(V)$; Eq. (1) makes this exact. The apparent choice between “route multiplicity” (a scalar) and “route connec-

tivity” (a structural object) dissolves, because whatever connectivity information the kernel carries enters robustness only through its projection onto $P(V)$.

The compression result gives that law an effective low dimensionality, and it is naturally read as a second stage of coarse graining: once the microscopic ensemble has been reduced exactly to $P(V)$, that distribution can, over a broad regime, be compressed further to only its first two moments while still reproducing the full robustness curve to the registered accuracy. Robustness is then, to that tolerance, a two-parameter function of the visiting statistics. Where second-order compression fails, the fail-

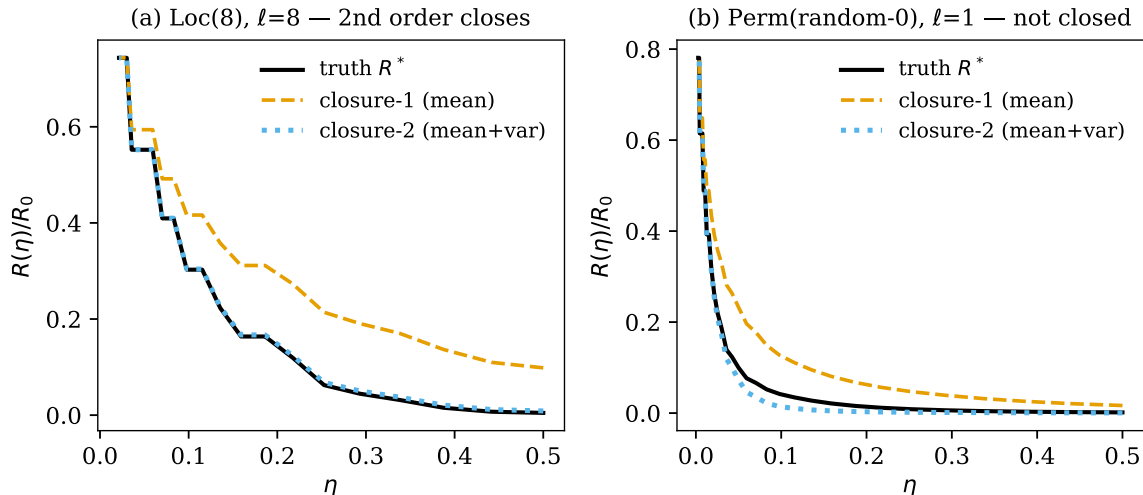


FIG. 3. Representative robustness curves $R(\eta)/R_0$ with the mean-only (closure-1) and mean+variance (closure-2) reconstructions and the truth R^* . (a) Loc(8) at $\ell = 8$: closure-2 lies on the truth (second order closes). (b) Perm(random-0) at $\ell = 1$: closure-2 improves on closure-1 but does not reach the truth (not closed). These two panels instantiate the blue and orange regions of Fig. 2.

ures are localized to coarse-resolution random permutations and sparse block kernels, which also exhibit broad visited-range distributions; whether that breadth is what causes the failure is not established here. That the mean is nonetheless insufficient across most tested regimes—the process almost always visits a spread of ranges, not a sharp one—keeps the two-moment reduction from being trivial.

The localization of the second-order failure suggests a separate question, which we do not address here and which lies outside the present preregistration: which features of $P(V)$ control the breakdown of moment compression? Answering it would require a new registered predictor (for instance a tail or skewness statistic of the visiting law) and is left to future work. Extending Eq. (1) beyond feed-forward dynamics—to recurrent chains, where a trajectory may revisit a region—is a second open direction.

VIII. CONCLUSION

On a matched family of feed-forward reactive chains, extensive-lesion robustness in dead-end mode is an exact functional of the visited-range distribution $P(V)$. The question of which information governs robustness thereby becomes a compression question, and its answer is a clean regime map: the mean is insufficient across most of the tested regimes, the mean and variance close the robustness curve to one percent in the large majority of cases, and higher-order distributional information is needed only in a localized set of cases, specifically coarse-resolution random permutations and sparse block kernels. The result was fixed before any outcome was computed and independently reproduced.

ACKNOWLEDGMENTS

The registered protocol, frozen model specification, analysis code, and a SHA-256 manifest of all frozen outputs are available so that the pipeline can be re-run and independently re-implemented.

Appendix A: Registered family

All kernels act on $n = 256$ positions with periodic boundaries and are exactly doubly stochastic by symmetry (no numerical balancing is applied). A member is one time-homogeneous kernel used at every layer. The five menus give 27 members: Loc(σ), the symmetric nearest-range walk with $P^R(x, y) = 1/(2\sigma + 1)$ for $|x - y| \leq \sigma$, $\sigma \in \{1, 2, 4, 8, 16, 32\}$ (6); Perm(π), the identity, cyclic shifts by $s \in \{1, 4, 16, 64\}$, and three fixed random permutations (8); Mix(c) = $(1 - c)\mathcal{K} + cU$ with U the uniform kernel, $c \in \{0.02, 0.05, 0.1, 0.2, 0.5\}$ (5); Block(M, c), M equal blocks with within-block uniform mixing at probability $1 - c$ and a uniform jump to another block at probability c , $M \in \{4, 16, 64\}$, $c \in \{0.02, 0.1\}$ (6); and ShiftMix(c) = $(1 - c)\text{shift}(16) + cU$, $c \in \{0.05, 0.2\}$ (2). The common parameters are $T = 64$, $\lambda = 1 - 2^{-1/64} \approx 1.077 \times 10^{-2}$ so that $R_0 = 1/2$, and $\ell \in \{1, 2, 4, 8, 16\}$. Random permutations use fixed generator seeds; the disorder-integrated estimator (Appendix C) uses a per-member trajectory seed, and the validation lesions use a separate removal seed, all registered before computation.

Appendix B: Central identity and redirect degeneration

Derivation of Eq. (1). In dead-end mode a lesion D deletes all transitions into removed blocks; a trajectory contributes to the survival iff every one of its positions lies in a surviving block, i.e. iff its set of visited blocks $\mathcal{V}(\omega)$ satisfies $\mathcal{V}(\omega) \cap D = \emptyset$. Because the committor of the embedded chain is flat within layers, the reactive weight of a surviving trajectory is unchanged by the lesion, so $R_{\text{dead}}(D)/R_0 = \mathbb{P}_\omega[\mathcal{V}(\omega) \cap D = \emptyset]$. For a fixed visited set of size $V = |\mathcal{V}(\omega)|$, a uniformly random removal set of r blocks avoids it with probability $\binom{n_b - V}{r} / \binom{n_b}{r}$. Averaging over D and then over ω gives Eq. (1). The disorder average is thus exact and analytic; only the trajectory expectation remains, and it depends on the law of V alone.

Single-edge exactness. As a code check, the within-layer flatness gives closed forms for the change ΔR under a single removed edge ($x \rightarrow y$) at one layer: in redirect mode $\Delta R = 0$ unless the row dies ($\Delta R = -R_0/n$), and in dead-end mode $\Delta R = -R_0 P^R(x, y)/n$. Both match a direct backward recursion to 10^{-12} .

Redirect degeneration. In redirect mode the deleted rows are renormalized over surviving successors. If no row is fully removed (no dead row), the renormalized operator is row-stochastic on S , which preserves the within-layer flatness, and $R_{\text{redirect}}(D)/R_0 = |S|/n$ independent of the kernel. A dead row requires $\text{supp } P^R(x, \cdot) \subseteq D$, impossible when $P^R(x, x) > 0$; 20 of the 27 members have strictly positive diagonal, so for them redirect carries no family information and $\eta_{1/2} = 1/2$ exactly. This is why dead-end is the informative intervention.

Toy check. Equation (1) and both degenerations are verified by exhaustive enumeration on a small chain ($n = 8$, $T = 4$, all trajectories and all removal sets), agreeing to machine precision.

Appendix C: Exact $P(V)$ and the estimator

For the 13 oracle members the law $P(V)$ is exact. Permutations are deterministic: each of the n starts yields one orbit, and $P(V)$ is the histogram of its distinct-block count. For $\text{Mix}(c)$ a step stays with probability $1 - c$ and otherwise jumps to a uniform position, so the visited set is the start together with $K \sim \text{Binomial}(T, c)$ uniform landings; V is the number of occupied cells among $m = K + 1$ balls in n_b cells [8]. We compute the occupancy law by the stable forward recursion

$$q_m(v) = q_{m-1}(v) \frac{v}{n_b} + q_{m-1}(v-1) \frac{n_b - v + 1}{n_b}, \quad q_0(0) = 1, \quad (\text{C1})$$

and $P(V) = \sum_k \text{Binomial}(T, k; c) q_{k+1}$. The inclusion-exclusion form of the same law cancels catastrophically for large m ; the oracle validation (Sec. VI) caught exactly this and the recursion above is the fix.

For the remaining members the truth is the disorder-integrated estimator $\hat{R}(\eta)/R_0 = M^{-1} \sum_{i=1}^M w(V_i, r(\eta))$ with $M = 10^6$ trajectories, which removes removal-set noise by construction (the average over D is done in w , not by sampling D). Because $V \in \{1, \dots, n_b\}$ the estimator is a bounded functional and its pointwise standard error is at most $1/(2\sqrt{M}) = 5 \times 10^{-4}$; a single trajectory sample serves all resolutions since $V(\ell)$ is read off the same positions.

Appendix D: Why maximum entropy, and the decision rule

Closures. Under the specified moment constraints, maximum entropy provides a canonical least-informative reconstruction of $P(V)$ [6, 7]: maximizing $-\sum_v p(v) \ln p(v)$ on $v \in \{1, \dots, n_b\}$ subject to a fixed mean gives $p_1(v) \propto e^{\lambda_1 v}$, and subject to a fixed mean and variance gives $p_2(v) \propto e^{\lambda_1 v + \lambda_2 v^2}$, the Lagrange multipliers being fixed by the exact $\mathbb{E}[V]$ and $\text{Var}(V)$. Using the same variational principle at both moment orders ensures that the two closures differ only in the information supplied—the mean alone, or the mean together with the variance. Within this common closure framework the reduction in curve error therefore measures the gain obtained by supplying the additional moment, which makes “how many moments are needed” a controlled comparison within a common closure family rather than an artifact of the reconstruction. The multipliers exist and are unique whenever the target moments lie strictly inside the moment space of the support; the degenerate cases ($\text{Var}(V) = 0$, or $\mathbb{E}[V]$ at a support endpoint) reduce to a point mass, and both were checked to be well posed for every member.

Decision rule. Curves are compared on a geometric grid in $u = -\ln(1 - \eta)$, $u_k = \ln 2 \cdot 2^{-k/4}$, whose endpoints bracket the analytic collapse limits $\eta_{1/2} \in \{1 - 2^{-1/n_b}, 1/2\}$; the registered linear-interpolation bound on R is 5×10^{-3} (not a bound on $\eta_{1/2}$). The statistic $D_{k,j}(\ell)$ of Sec. IV is evaluated on grid points, so it does not depend on interpolation. With $\tau = 10^{-2}$, a member’s exceedance count is $m_j = \#\{\ell : D_{1,j}(\ell) > \tau\}$, and the registered *existential* family rule is: H_0 if $m_j = 0$ for all members; H_{dist} if $m_j \geq 2$ for some member; inconclusive otherwise. This matches the universal reading of coordinate sufficiency: the mean is a sufficient coordinate for the family only if it reproduces every member’s curve. Within H_{dist} , an exceeding cell is classified “second order closes” if $D_{2,j}(\ell) \leq \tau$, and otherwise “higher” if $D_{2,j} \geq D_{1,j}/2$ (second order does not substantially improve) or “2nd improves” if it does. Because τ exceeds the estimator error budget ($\leq 2 \times 10^{-3}$ in sup norm) five-fold, and every kernel with concentrated V is an exact- $P(V)$ oracle member, estimation noise cannot manufacture the family verdict. Table I lists the per-member outcomes.

Preregistration and audit. The protocol was drafted,

TABLE I. Per-member outcome by resolution. $\circ$ mean sufficient; $\bullet$ mean+variance closes to $\tau = 10^{-2}$; ∇ second order improves but does not close; $\blacktriangle$ higher-order. m_j is the number of resolutions at which the mean fails.

member	$\ell=1$	$\ell=2$	$\ell=4$	$\ell=8$	$\ell=16$	m_j
Loc(1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Loc(2)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Loc(4)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Loc(8)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Loc(16)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Loc(32)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Perm(identity)	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	0
Perm(shift-1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Perm(shift-4)	$\bullet$	$\bullet$	$\circ$	$\circ$	$\circ$	2
Perm(shift-16)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\circ$	4
Perm(shift-64)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Perm(random-0)	∇	∇	∇	$\circ$	$\bullet$	4
Perm(random-1)	∇	∇	∇	$\circ$	$\bullet$	4
Perm(random-2)	∇	∇	∇	$\blacktriangle$	$\bullet$	5
Mix(0.02)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Mix(0.05)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Mix(0.1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Mix(0.2)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Mix(0.5)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Block(4,0.02)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Block(4,0.1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Block(16,0.02)	$\bullet$	∇	∇	∇	$\bullet$	5
Block(16,0.1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
Block(64,0.02)	∇	∇	$\bullet$	$\bullet$	$\bullet$	5
Block(64,0.1)	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	5
ShiftMix(0.05)	$\bullet$	$\bullet$	$\bullet$	$\circ$	$\circ$	3
ShiftMix(0.2)	$\bullet$	$\bullet$	$\bullet$	$\circ$	$\circ$	3

adversarially reviewed, and hash-frozen before any robustness curve was computed. An earlier version was *superseded before any outcome was computed*, when two of its registered predictors were found degenerate on the unperturbed process alone—a normalized maximum-flow that is identically unity by conservation of reactive mass, and an itinerary entropy with no closed form on part of the family—and when the redirect intervention was shown analytically to fix the endpoint on most members. The question and the endpoints were then rebuilt from the exact identity Eq. (1), still before outcomes, and re-frozen. The full pipeline was then re-implemented independently from the frozen family specification and agreed on the moments and curves within the registered tolerances. A SHA-256 manifest of the frozen protocol, code, and outputs accompanies the paper.

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